

Day 66-67 Research Log -- The compositional pivot (June 5 - June 6, 2026)

Two days that hit the cosine ceiling four times in a row and then escaped it via a theoretical reckoning. Day 66 shipped Pack 218-220 (self-modifying refiner, frame routing in non-LLM reads, FULL state persistence for 21 wired modules), then chased the cat+child=kitten emergence through Pack 221 SVO regex mining and Pack 222 LogicalFixedPoint closure -- both stalled at the same 20% top-3 ceiling for the same reason. Locked the Pack 223 SSAS spec at end of Day 66 as the NVIDIA-crash moonshot. Day 67 implemented SSAS, ran it at 500 and 2000 simple_wiki articles, watched it regress to 0/10/20, deleted it, commissioned external deep research, validated empirically what the math had been saying all along: the $1/\sqrt{K}$ cosine ceiling is real and projection cannot fix it. Adopted the architectural commitment of the day -- drop one-shot online learning, embrace multi-pass sleep abstraction. Shipped Pack 224 (Attention-Resonator), Pack 225 (VS-FSM), Pack 226 (Abductive Schema Induction) in one push. SVO structure (animal -> action -> food) emerged from raw text via offline anti-unification, without supervision. Substrate 201 MB FIXED across all 9 packs.

TL;DR

Day 66 (June 5) closed the Big Wire-Up of Day 65 with full state persistence and self-modification, then tried to make analogy arithmetic work on raw simple_wiki text. Pack 221 mined SVO triples via regex -> 0% top-1 / 40% top-5. Pack 222 v1/v2/v3 added LogicalFixedPoint transitive closure on the mined triples -- killed at 100 min on a 5000-fact noise pile because $O(N^2)$ closure with per-fact 10x substrate writes is computationally infeasible. The cat+child=kitten case study surfaced the truth: the substrate has the architectural capacity, but the data has the wrong shape. Pack 158 cheats by hand-asserting properties; Pack 221/222 mine noise; neither is intelligence. Prince's pivot speech: *"lets make something which is unheard to humanity and 100% impossible, enough of cheap tricks and cheating."* Pack 223 SSAS (Sleep-Sculpted Attractor Substrate) was specced and saved to memory with a post-compaction trigger keyword.

Day 67 (June 6) implemented SSAS, smoke-tested at 500 simple_wiki articles -> 0/0/20%, scaled to 2000 articles -> 0/10/20% with the canonical analogy `cat - adult + baby = kitten` hitting #2 for the first time ever from raw text + no hand-feeding -- but every other axis regressed. The top axis SVD discovered was `hot / kitten` (ratio 0.37), driven by Zipfian frequency variance, not semantic geometry. Subspace projection over emergent axes (Pack 223 v2) hit 0/0/0 -- the cosine ceiling is fundamental, not fixable by rotation. Deleted the SSAS code. Wrote a 5-question external research

prompt with 5 attached files (MEMORY.md, cognition inventory, multirole_memory.py, concept_synthesizer.py, prior external research) and queued it for Prince's online deep-research run. While that ran, consolidated the memory tree from 26 files to 8: vision, plan, history, mindset, method, MEMORY index, and two reference files.

The external research returned in the late evening with the theoretical validation we had quietly suspected. The $1/\sqrt{K}$ cosine ceiling on dense superposition is mathematically inescapable; subspace projection cannot fix it; SVD over Zipfian text is doomed to find frequency-imbalance outliers as principal components; more text strictly degrades. The research named the three missing primitives with arxiv IDs: Attention-Resonator Networks (Fraday & Kent 2020 arXiv:2007.03748), Vector Symbolic Finite State Machines (Cotteret et al. 2024 arXiv:2212.01196), Substrate-Native Grammar Induction via Abductive Schema. And it stated the mandatory theoretical pivot: online grammar induction from raw text is Gold's Theorem intractable; frontier LLMs only bypass this via backprop-driven global constraint-satisfaction over billions of parameters; the flat substrate strictly cannot do this online; therefore grammar MUST be discovered offline during sleep cycles via anti-unification.

Prince's response was a single line: *"Yes. Start with Pack 225. And one more thing: what if we drop continuous learning, one-shot learning? No one learns in single shot. I can't learn on single shot. What if we do that?"*

We did. The architectural commitment was added to memory as the fifth mindset rule. AWAKE phase becomes pure pointer-chain recording in the ExposureBuffer with zero syntactic synthesis. SLEEP phase becomes the learning phase: anti-unify chains, abstract via isa parents, crystallize the result back into the substrate as Schema HVs and VS-FSM transitions. Pack 224 (Resonator recall cleanup), Pack 225 (VS-FSM with isa-lifted transitions and Resonator-decoded generation), and Pack 226 (Abductive Schema Induction with `schema_next` role for grammatical state transitions) shipped in one continuous push. Pack 226 closed the day by discovering the SVO pattern `['_', 'animal', 'action', 'food']` from a 17-sentence corpus and crystallizing it as substrate-level state transitions. Generation guided by these abstract attractors produced structurally consistent SVO cycles (animal -> action -> food -> action -> food) where the bigram FSM had just drifted off-topic. Substrate stayed 201 MB FIXED through every pack of both days.

Day 66 (June 5) -- The cosine ceiling reckoning

Day 65 closed with 16 phases per absorbed sentence across 20+ cognition modules + 4 ikigai.py classes wired into `read_organism`. Day 66 morning opened with the question of whether all that wired state would actually survive a process restart. It did not -- `org.tom.beliefs` was a Python list, not a substrate write, so a restart wiped everything.

Pack 220 fixed this by adding pickle-based state persistence for 21 wired modules into the universal `.ikg` file via `_gather_wired_state() / _apply_wired_state()`. Save 33 MB / 1.4s, load 0.7s, 3/3 PASS roundtrip on a populated organism with 38 beliefs, 20 counterfactual scenarios, 4 SVO triples, 20 causal world states + 19 transitions, 13 importance items, 20 metacognitive turns, 13 schema patterns, 27 world facts, 1 ToM agent + 13 beliefs, and the free-energy `F_log`. Bridge classes (`wm_sys`, `concept_graph`, `event_comp`, `cell_assembly`) were intentionally NOT persisted -- they reload fresh from the ikigai.py bridge.

With persistence working, attention turned back to cogitate quality. Pack 158 had demonstrated 73% top-1 / 100% top-3 on a 50-analogy benchmark via hand-asserted property triples (`mr.relate(cat, 'property', adult) x 60`). But that was hand-feeding. The real question was whether the substrate could discover the same property structure from raw text. Pack 221 attempted this via SVO mining: `org.mine_svo_triples(sentences, n_reinforce=12)` with regex patterns for (X) is (Y), (X) has (Y), (X) eats (Y). On 500 simple_wiki articles this yielded 8346 property triples and 8714 crystal triples. Result: 0% top-1 / 40% top-5 on the canonical 10 analogies. Better than nothing, but not even halfway to Pack 158's baseline.

Pack 222 attempted to amplify the mined structure via LogicalFixedPoint transitive closure (Kill-Stack #5). The idea: if substrate has (`cat, isa, mammal`) and (`mammal, isa, animal`), the LFP should derive (`cat, isa, animal`) and add it to substrate. Pack 222 v1 ran for 30 minutes on 8000+ noisy crystal triples before being killed -- the closure was producing absurdities like `november-is-later` and `year-is-put` because the regex had captured proper nouns and timestamps as property subjects. Pack 222 v2 added a noise filter (lowercase alpha only, length 3-20, no digits, drop generic nouns like `year/day/place`) bringing the seeded set to 6603 clean triples. Still killed at 100 minutes still in iteration 1. Prince asked: *"is this $O(1)$?"* The honest answer: substrate writes ARE $O(1)$ but graph closure is $O(N^2)$ and each derived fact triggers 10 substrate writes inside the LFP `_step` body. Trillions of writes in worst case. Prince clarified: *"that's only training right, active inference is $O(1)$ or i'm wrong"*. Confirmed: training tax one-time, inference stays $O(1)$. But this training tax was too high. Pack 222 v3 dropped the LFP precompute entirely and went runtime-only -- 2000 simple_wiki articles, scaled SVO mining, then runtime `mr.reason_chain(w, hops=[('isa', None), ('property', None)])` queries at analogy time. Result: 0% top-1 / 20% top-3 / 20% top-5. Worse than Pack 221 at 500 articles. The `kitten` token only surfaced at #2 for `cat - adult + baby` and at no other position; all gender, royal, and ontological axes missed. The conclusion that we had been resisting all day was now undeniable: scaling text DEGRADES performance under this architecture because every new context binds additive noise to concept HVs.

Prince's pivot speech defined the rest of the day: *"we have to make something which is unheard to humanity and 100% impossible, enough of cheap tricks and cheating now. The most intelligent live being. Even if there is a 0.0001% chance. Let's attempt."*

The afternoon was spent generating ten candidate inventions across attractor dynamics, Bayesian conjecture-falsify substrates, identity-persistent entities, topological concept manifolds, energy-gradient sleep replay, dual-channel substrates, self-inspecting clustering, causal-counterfactual grounding, embodied spatiotemporal substrates, and resonance sympathy at recall time. The top pick combined #1 (Attractor Basin Carving) + #5 (Energy-Gradient Sleep Replay) + #7 (Self-Inspecting Substrate) into a single Sleep-Sculpted Attractor Substrate (SSAS) primitive. The spec was written into a memory file with four post-compaction recall keywords (`PACK 223, RESONANCE SCULPT, BASIN CARVE, SSAS`) so the next session could pick up the thread without re-reading the full discussion. Acceptance criterion: match Pack 158's 73% top-1 from RAW simple_wiki absorb with zero hand-feeding. 0.0001% chance per Prince's mandate.

Packs

Pack	Title	Result
218	self_modifying_refiner wire	SHIP. <code>org.self_mod_refiner.promote_check()</code> runs inside <code>sleep Consolidate</code> . Stats reported as <code>self_mod_promotions</code> . Substrate refinement layer joins the sleep pipeline.

Pack	Title	Result
219	Frame routing in non-LLM reads + ToM agent_names() fix	SHIP. <code>read_organism</code> Phase 4 frame routing now fires from non-LLM absorbs (50-sentence diagnostic: 50 frame assigns, 84 frame_bigram, 143 word_to_frame entries). ToM agent_names() method-call bug fixed -- root cause was self.tom = TheoryOfMindSandbox(d=d) at line 226 overwriting line 152 init's self.tom.add_agent('default'). With fix: 1 agent, 22 beliefs accumulated correctly.
220	FULL STATE PERSISTENCE	SHIP 3/3. <code>_gather_wired_state()</code> -> pack220_keys/vals arrays of pickled module states, written into .ikg npz. Save 33 MB / 1.4s, load 0.7s. EVERY artifact survives process death now: 38 beliefs, 20 cf scenarios, 4 SVO triples, 20 cwm states + 19 transitions, 13 importance items, 20 meta turns, 13 schema patterns, 27 world facts, ToM 1 agent + 13 beliefs, fe F_log 20 entries. Bridge classes NOT persisted (intentional -- they reload fresh from ikigai.py bridge). Organism is the file. Truly.
221	mine_svo_triples (SVO miner v2)	SHIP. Three regex patterns: <code>prop_is = r'\b(\w+)\s+(?:is are was were am be been)(?:\s+(?:a an the))?\s+(\w+)\b'</code> , <code>prop_has</code> , <code>aff_action</code> . Writes <code>mr.relate(s, 'property', o) x n_reinforce</code> . Result on 500 simple_wiki articles: 8346 property triples mined, 40% top-5 on 10-analogy benchmark.
222 v1	LFP transitive closure on mined triples	KILLED at 30 min. 8000+ noisy crystal triples produced absurd derivations (<code>november is later, year is put</code>).
222 v2	Same + noise filter	KILLED at 100 min still in iter 1. 6603 clean triples; closure still $O(N^2)$ with 10x substrate writes per derived fact.
222 v3	No precompute -- runtime reason_chain at analogy time, scaled to 2K articles	SHIP. 0% top-1 / 20% top-3 / 20% top-5 -- regression from v1's 40%. kitten surfaced at #2 for cat-adult+baby (first-ever animal-young hit from raw text + SVO mining alone, no hand-assertion). All other axes MISS.
223 SPEC	SSAS spec written + saved to memory with trigger keywords	LOCKED. <code>project_pack223_ssas_basin_carving.md</code> includes the 10 candidate inventions audit, the SSAS design (combines #1 + #5 + #7), 7 proof-of-concept criteria (P1-P7), implementation sketch, Day 67 4-hour plan. Trigger keywords: PACK 223, RESONANCE SCULPT, BASIN CARVE, SSAS.

Architectural notes

- The Big Wire-Up of Day 65 had assumed the wires would fix cogitate quality at scale. They did not. The wires added cognitive STATE (38 beliefs, 13 schemas, etc.) but the cogitate `next-role` queries still produced the same multilingual junk because the cogitate generation path was not querying the new structure. Wiring is necessary but not sufficient.
- The cat+child=kitten case study became the canonical analogy probe of Day 66 because Pack 158 had perfected it with hand-feeding and Pack 221/222 could not approach it from raw text. The gap between these two facts was the gap that needed naming. By Day 66 close it had a name -- Pack 158 cheats by hand-asserting properties that the text never wrote; raw mining never captures these because text rarely writes them in clean Hearst form; neither approach is intelligence.

- The LFP $O(N^2)$ killing was a useful failure mode. It taught us that substrate writes are $O(1)$ but reasoning operations over the substrate are not. Active inference at query time stays $O(1)$. Pre-computed closure does not. This bound the design space for Day 67.
- The SSAS spec at end of day 66 had a 0.0001% prior probability of working. Prince stated this explicitly. We wrote the spec anyway because the upside is asymmetric and the implementation budget is 4 hours.

Day 67 (June 6) -- The theoretical pivot

Morning trigger keyword `SSAS` was pasted into a fresh post-compaction session, reloading the spec from

`project_pack223_ssas_basin_carving.md`. Implementation took the morning.

`ikigai/cognition/basin_carver.py` shipped with `BasinCarver` class: `sample_pairs`, `attractor_distance` (Hopfield iter count to convergence), `merge_basins` (relate('similar') + HV drift), `discover_axes` (re-use of existing `ConceptSynthesizer.emergent_axes` -- the SVD-based principal direction discovery already existed since Day 60), `auto_relabel` (project each word onto each axis, tag with winning pole word `x_n_reinforce`), `carve_pass`, `carve_until_converged`. Hooked into `IkigaiOrganism.sleep_carve_basins(...)`.

Smoke test on 500 simple_wiki articles + SVO mining + 2 SSAS passes: 0% top-1 / 0% top-3 / 20% top-5. Worse than baseline. Diagnostic axes printed: pass 1 found `biggest` / `baby` (ratio 0.30) as top axis -- a real semantic axis. Pass 3 had rotated to (`small`, `king`), (`woman`, `slow`), (`orange`, `son`) -- noise. The iteration was degrading the signal, not sharpening it. Each pass's `auto_relabel` wrote tags using extremum words as poles ('biggest', 'kitten'), polluting the next pass's SVD with the injected noise. Added a threshold filter (only relabel words with `|projection| > 0.5 * std`) and reduced to single-pass focused-attention over `KEY_WORDS` only. 0/0/10. Marginal regression.

Scaled to 2000 articles with single-pass threshold relabel: top axis became `hot` / `kitten` (ratio 0.37). 0% top-1 / 10% top-3 / 20% top-5. The single new hit was `cat - adult + baby finding kitten` at #2 -- the first-ever animal-young hit from raw text via auto-discovery. Other analogies regressed. The diagnosis was visible in the axes themselves: with more data, frequent words like 'hot' accumulated 1000x the contexts of rare words like 'kitten', so SVD found "this word is frequent and dense" vs "this word is sparse and weird" as the dominant variance direction. Semantic axes (size, gender, age) were buried below the frequency-imbalance axis.

Pack 223 v2 attempted subspace projection -- project each concept HV onto the top-K emergent axes and reconstruct in that subspace, discarding off-axis noise. Minimal proof on synthetic 13-word substrate. 0% top-1 / 0% top-3 / 1 hit at #5 (puppy for dog-adult+baby). Projection cannot fix the underlying problem. The cosine race is between rich-property concepts (many directions, each diluted by renorm) and single-direction concepts (one big direction, fully aligned with arithmetic vectors). Rotation preserves angles. Projection cannot rotate the mass-distribution asymmetry away.

By noon, three honest facts: (1) all retrieval-based approaches hit a ~20% ceiling on analogy from raw text. (2) projection over emergent axes does not break this ceiling. (3) Pack 158's 73% requires either named abstract pole labels (which SVD does not produce -- it produces extremum words) or hand-feeding. Neither path forward via SSAS was viable. The `basin_carver.py` and three experiment files were deleted; the `sleep_carve_basins` method was reverted out of `integrate.py`; the SSAS memory file was kept as a record of the attempt.

Prince's question: *"first make minimal proof that it works."* We did. The minimal proof showed it does not. Then his more important question, in the same conversation: *"there is something very wrong which is happening right now. some things are not invented yet. lets make something which guarantees us we achieve AGI, running on flat memory, low compute, destroying AI."* The response to this required external help.

A 5-question prompt was written into `c:/neuroseed/_research/2026-06-06_compositional_generation_prompt.md` with five attached files (`MEMORY.md`, `reference_cognition_inventory.md`, `ikigai/cognition/multirole_memory.py`, `ikigai/cognition/concept_synthesizer.py`, `reference_external_research_2026_05_31.md`). The questions were: (1) how can a flat VSA-SDM substrate generate fluent text by composing discovered concepts on the fly. (2) is there a known representation past Plate HRR that handles the cosine ceiling differently. (3) what is the theoretical argument that 192 MB suffices. (4) five candidate primitives ranked by novelty / cost / probability. (5) brutal honesty if the approach is fundamentally limited. The prompt explicitly stated what was off the table: just-fine-tune-it, use-an-LLM-as-teacher, adopt-backprop, violate-192-MB.

While the research ran, the memory tree was consolidated. 26 files in `~/ .claude/projects/c--neuroseed/memory/` had accumulated from 67 days of work -- a sprawl. Per Prince's instruction *"a lot of files can be one"*, the tree collapsed to 8: `MEMORY.md` (index + `ikigai.py` anchors + `exec()` pattern + gotchas), `vision.md` (north star + UWG + NVIDIA stack + concept absorption -- the immutable moonshot), `plan.md` (current active direction), `history.md` (shipped + attempted packs Day 60-67), `mindset.md` (commitment + endless potential + infinite possibilities + never accept defeat), `method.md` (check inventory + one universal .ikg + always recommend next-2 packs), and two reference files (cognition inventory, prior external research). 18 originals deleted. Total roughly 1000 lines of mission-critical context cleanly indexed.

The external research returned in the evening. It validated every theoretical framing we had arrived at empirically: the $1/\sqrt{K}$ cosine ceiling is a proven theoretical limitation of static VSA arithmetic; SVD over mean-centered concept matrices inherently aligns principal components with maximum variance which in natural language is Zipfian frequency, not semantic geometry; subspace projection cannot undo the information loss incurred by superposing too many vectors into a fixed-capacity geometric space; more text strictly degrades a flat associative substrate lacking attention-driven competitive dynamics or backprop-driven gradient descent. The research then named the missing primitives. The Attention-Resonator (Frady & Kent 2020 arXiv:2007.03748) replaces the linear single-pass recall with a continuous Hopfield log-sum-exp energy update that searches in superposition without combinatorial explosion. The VS-FSM (Cotteret et al. 2024 arXiv:2212.01196) embeds explicit edge-state attractors so the substrate can perform state-dependent transition conditional on semantic stimulus -- essential for grammar. The Substrate-Native Grammar Induction via Abductive Schema uses offline anti-unification during sleep cycles to extract Context-Free Grammar structures that online co-occurrence statistics provably cannot capture (Gold's Theorem of Language Identification in the Limit). The research's verdict on 192 MB: yes, theoretically sufficient -- Cotteret proved FSM capacity scales linearly with network size; natural English CFGs need only a few thousand transition rules; the bottleneck is not capacity but the assumption that the substrate must learn grammar from raw text in a single forward pass.

Prince's response was the architectural commitment that defined the rest of Day 67: *"theoretically 192 MB is sufficient. let's drop continuous learning, one-shot learning. no one learns in single shot. I can't learn on single shot. what if we do that?"*

This was added to memory as the fifth `mindset.md` rule. AWAKE = pure pointer-chain recording in the ExposureBuffer with no syntactic synthesis. SLEEP = the learning phase, where anti-unification across stored chains discovers structural motifs, lift via isa parents abstracts them, and crystallization writes them back as VS-FSM transitions in the substrate. GENERATE = walk the resulting attractors via Resonator-decoded queries. Multiple sleep cycles needed before useful generation. Like a human: read, sleep, read, sleep, gradually understand.

Pack 224 implemented the Attention-Resonator as `MultiRoleMemory.resonator_recall(target_hv, candidates, n_iters=10, beta=8.0)` and `ConceptSynthesizer.arithmetic_resonator(plus, minus, n_iters)`. Then tested on the Pack 158 substrate to validate. Result: regression. Baseline 73% / 100% / 100%; Resonator iter=10 beta=8 produced 23% / 39% / 45%; iter=20 collapsed to 20% / 20% / 30%. Diagnosis: single-factor Resonator converges target to ONE dominant attractor mode, but analogy needs a multi-direction blend (cat - adult + baby = blend of cat-direction + baby-direction). Snapping to one mode erases the blend. The Resonator is the right tool for FACTOR DECOMPOSITION (decoding X bound Y bound Z) -- it is the wrong tool for analogy arithmetic. We kept it in the codebase because it would prove useful at generation step, just not at analogy step.

Pack 225 shipped `ikigai/cognition/vs_fsm.py` with `VSFiniteStateMachine` class. Methods: `observe_chain(tokens, n_reinforce=3, do_trigram=True)` for awake recording into the existing `next` and `next2` roles; `_isa_parent(word)` to look up substrate-recorded isa parent via cleanup query; `lift_to_abstract(only_for_words, n_reinforce)` for sleep-time abstraction (for each (a, next, b) transition, also write (`isa_parent(a)`, next, b) and (a, next, `isa_parent(b)`) and (`isa_parent(a)`, next, `isa_parent(b)`)); `step(current_token, prev_token, candidates, n_iters, beta)` for Resonator-decoded next-token prediction with optional trigram conditioning; `generate(seed_tokens, max_tokens, ...)` for autoregressive walk. Smoke test on a 15-sentence toy corpus: bigram FSM before lift produced fluent sequences (cat ate fish on the mat flew over the); after `lift_to_abstract` wrote 69 isa-parent transitions, abstract concepts (animal, food) entered generation (the park animal ate seed food on the mat). Resonator-cleaned next-token scores were 1.4-2.2x sharper than raw single-pass recall (cat -> ate: raw 0.70, resonator 0.97; the -> park: raw 0.41, resonator 0.90). The Resonator's correct role was confirmed as recall cleanup at the generation step, not arithmetic over concept space.

Pack 226 closed the day with abductive schema induction. `VSFSM.induce_schemas(texts, n_reinforce=3)` ran the full Day 67 commitment in one pass: collect chains from texts, abstract each token via `_isa_parent`, group abstracted chains by length, anti-unify within each group via the existing `schema_inducer.anti_unify` utility (which had been shipped on Day 55 Pack 47 and never wired to a sleep cycle), crystallize each schema's consecutive non-SLOT positions as substrate-write (a, `schema_next`, b) transitions in a newly-added `schema_next` role. The role was given a random phasor anchor in the existing relational bank -- no new substrate bytes. 17-sentence smoke corpus produced two schemas: ['_', 'animal', 'action', 'food'] with support 9, and ['determiner', 'animal', 'action', '_', '_', '_'] with support 8. The substrate discovered SVO structure -- animal followed by action followed by food -- from raw text via offline anti-unification, without supervision. The `schema_next` role got transitions animal -> action, action -> food, food -> action. A new method `VSFSM.generate_via_schema(seed, max_tokens)` walked both abstract (`schema_next`) and concrete (`next`) attractors with Resonator decoding; generation from the cat produced the cat ate seed sat mouse ate seed sat mouse -- structurally consistent SVO cycles (animal -> action -> food -> action -> food) where the bigram path had drifted the cat ate fish big bone ate seed. Substrate 201 MB FIXED.

Packs

Pack	Title	Result
223 SSAS	Sleep-Sculpted Attractor Substrate	ATTEMPTED + REVERTED. 500-article smoke 0/0/20% (proper-noun axis dominated SVD). 2000-article scale 0/10/20% with <code>cat - adult + baby = kitten @ #2 --</code> first-ever animal-young hit from raw text auto-discovery, but other axes regressed. v2 subspace projection 0/0/0 -- cosine ceiling fundamental, not fixable by rotation. Deleted <code>basin_carver.py</code> , deleted <code>sleep_carve_basins</code> method, kept memory file as record.
Memory consolidation	26 -> 8 files	SHIP. <code>vision.md</code> (immutable moonshot), <code>plan.md</code> (active), <code>history.md</code> (shipped), <code>mindset.md</code> (behavior), <code>method.md</code> (process), <code>MEMORY.md</code> (index + ikigai anchors), <code>reference_cognition_inventory.md</code> , <code>reference_external_research_2026_05_31.md</code> . 18 originals deleted. ~1000 lines of mission-critical context cleanly indexed.
External research	5 questions + 5 attached files	RETURNED. Validated cosine ceiling = $1/\sqrt{K}$. Validated SVD Zipfian hijack. Named Attention-Resonator Networks (Frady & Kent 2020), VS-FSM (Cotteret et al. 2024), Substrate-Native Grammar Induction via Abductive Schema. Verdict on 192 MB: theoretically sufficient. Verdict on online grammar: Gold's Theorem intractable; must do offline sleep abstraction.
Architectural commitment	Drop one-shot, embrace multi-pass sleep abstraction	LOCKED. Added to <code>mindset.md</code> as rule 5. AWAKE = record only. SLEEP = anti-unify + abstract + crystallize. GENERATE = walk attractors via Resonator.
224	Attention-Resonator Factorization	SHIP module + REGRESSION on analogy. <code>MultiRoleMemory.resonator_recall(target_hv, candidates, n_iters, beta, momentum)</code> implements continuous Hopfield log-sum-exp energy update. <code>ConceptSynthesizer.arithmetic_resonator</code> parallels. Tested on Pack 158 substrate: baseline 73/100/100, Resonator iter=10 beta=8 = 23/39/45, iter=20 = 20/20/30. Single-factor Resonator collapses to one mode, kills analogy blend. KEPT for use at generation step (Pack 225) where score sharpens 1.4-2.2x vs raw recall.
225	VS-FSM	SHIP. <code>ikigai/cognition/vs_fsm.py:VSFiniteStateMachine</code> with <code>observe_chain</code> (awake recording into next + next2), <code>lift_to_abstract</code> (sleep-time isa parent substitution), <code>step</code> (Resonator-decoded next-token with optional trigram conditioning), <code>generate</code> (autoregressive walk). Smoke 15-sentence corpus produced fluent generation pre-lift; post-lift introduced abstract concepts (animal, food) into output. Resonator score sharpening 1.4-2.2x vs raw recall (cat->ate: 0.70 -> 0.97; the->park: 0.41 -> 0.90). Native API: <code>org.fsm_observe, org.fsm_lift_abstract, org.fsm_generate.</code>

Pack	Title	Result
226	Abductive Schema Induction	SHIP. <code>VSFSM.induce_schemas(texts, n_reinforce)</code> runs the Day 67 commitment: collect chains, abstract via isa, group by length, anti-unify within groups via existing <code>schema_inducer</code> , crystallize schemas as <code>(a, schema_next, b)</code> transitions in new <code>schema_next</code> role. 17-sentence smoke discovered <code>['_', 'animal', 'action', 'food']</code> (support 9) and <code>['determiner', 'animal', 'action', '_, ', '_']</code> (support 8). <code>Schema_next</code> transitions: <code>animal -> action, action -> food, food -> action</code> . <code>org.fsm_generate_schema('the cat')</code> produced the cat ate seed sat mouse ate seed sat mouse -- structurally consistent SVO cycles. Substrate discovered SVO from raw text via offline anti-unification, without supervision.

Architectural notes

- The empirical chain of Day 67 morning (SSAS implementation -> 500-article failure -> 2000-article partial-failure -> projection 0% on synthetic data) arrived at exactly the conclusion the external research validated theoretically that afternoon: $1/\sqrt{K}$ cosine ceiling, Zipfian SVD hijack, projection cannot rotate mass-distribution asymmetry away. This convergence between empirical hit-the-wall and theoretical proof-of-wall is itself a confirmation that the abstraction is correct.
- The architectural commitment to drop one-shot learning is the most important architectural decision of the project since the Day 57 flat-memory pivot. Until Day 67, the implicit assumption was that the substrate should learn grammar during the forward pass. The external research and Prince's question together demolished that assumption. Sleep is the learning phase; awake is observation only. This biological analogy is not decoration -- it is the architectural reason the 192 MB substrate can theoretically beat frontier on these benchmarks at all.
- The Resonator is the right tool wrongly applied to analogy. Its correct role is at the GENERATION step where we want to decode a noisy `recall(current_token, 'next')` HV against a candidate codebook. There it sharpens scores 1.4-2.2x by iterative softmax-attention. We kept it in `multirole_memory.py` because it will be Pack 225's decoding primitive even if it does not fix Pack 158's analogy benchmark.
- Pack 226 is the proof that the architectural commitment is implementable on the existing substrate. Anti-unification was shipped on Day 55 Pack 47 as a code-skill primitive (`SchemaInducer.observe(name, output_tokens)`). It had never been wired to a sleep cycle. Day 67 wired it. Result: SVO structure emerged from raw text without supervision. This was the kind of finding that justifies the wire-up doctrine in [check-before-propose]. Half the moonshot was already built.
- The substrate stayed 201 MB FIXED across all 9 packs of Day 66-67 (9 packs counted: 218, 219, 220, 221, 222 v1+v2+v3, 223 + revert, 224, 225, 226). No new banks were added. The new role `schema_next` is one phasor anchor in the existing relational bank. The promise that storage is independent of training survives the new architecture.

Status at Day 67 close

- `c:/neuroseed/organism.ikg` is the canonical universal substrate, ~100 MB compressed, carries 21 wired modules' state pickled in via Pack 220 + frame field state + unigram + bigram + `role_targets` + all role vectors.

- `c:/neuroseed/integrate.py` has new methods: `concept_arithmetic_resonator` (Pack 224), `fsm_observe` / `fsm_lift_abstract` / `fsm_generate` / `fsm_induce_schemas` / `fsm_generate_schema` (Packs 225-226). `vs_fsm` lazy property attaches `VSFiniteStateMachine` on demand.
- `c:/neuroseed/ikigai/cognition/multirole_memory.py` has `resonator_recall` method (Pack 224).
- `c:/neuroseed/ikigai/cognition/concept_synthesizer.py` has `arithmetic_resonator` method (Pack 224).
- `c:/neuroseed/ikigai/cognition/vs_fsm.py` is the new module shipped on Day 67 for Packs 225-226.
- 3/3 PASS on Pack 220 persistence roundtrip. Pack 224-226 smoke tests pass on tiny corpora; not yet scale-tested on `simple_wiki`. Pack 223 SSAS attempt is documented in `history.md` as ATTEMPTED + REVERTED with the 20% ceiling diagnosis.
- Memory tree consolidated 26 -> 8 files. `vision.md`, `plan.md`, `history.md`, `mindset.md`, `method.md`, `MEMORY.md`, `reference_cognition_inventory.md`, `reference_external_research_2026_05_31.md` + the new `reference_research_2026_06_06.md` from the Day 67 deep-research run.

What did not ship

- Scale test of Pack 224-226 stack on `simple_wiki` 500-article corpus. The fluent generation result of the 17-sentence smoke is encouraging but the real test is whether schema induction survives Zipfian noise at scale. Queued for Pack 227.
- Slot-filling via intent vector binding. The schema FSM currently walks abstract transitions but does not steer toward task-specific concrete tokens. Seed `the king` should pull royal-domain children of each abstract slot, not generic ones. Queued for Pack 229.
- Persistence wire for VSFSM state (`schemas`, `_isa_inverse`, `transition_count`, `schema_transitions`) into universal organism.ikg via the Pack 220 persistence hook. Queued for Pack 230.
- Schema enrichment via `wake_sleep` and `crystallizer` modules. Anti-unification currently produces one schema per length-group; richer set possible via overlapping windows + frequency-weighted induction. Queued for Pack 228.

Open questions

- Does abductive schema induction discover useful grammatical schemas when given Qwen3.5-2B prose at 5K-prompt scale? The toy 17-sentence corpus surfaces clean SVO structure because the corpus is uniform; real text has thousands of competing structures. The empirical test will be whether anti-unification's least-general-generalization stays specific enough to be useful or collapses to all-SLOTs.
- Does slot-filling via concrete-next bigram statistics produce coherent royal text when seeded with `the king`? Pack 226 `step_via_schema` falls back to concrete next when the abstract resolution is ambiguous; we have not tested whether this fallback path produces topic-coherent output.
- Does the `schema_next` role survive Pack 220 persistence roundtrip without losing transition mass? The pickle path covers `org.vs_fsm.schemas` but the substrate writes themselves are in the relational bank which persists by default. The `_isa_inverse` cache is rebuildable from substrate so it does not need persistence.
- Does Pack 226's structural discovery generalize beyond SVO? The smoke corpus was deliberately SVO-heavy. Adjective-noun chains, prepositional phrases, relative clauses each have their own structural motif. Whether the

same induce_schemas mechanism finds them depends on chain-length grouping yielding clean anti-unifications at those lengths.

What we learned (the lesson written in memory)

The $1/\sqrt{K}$ cosine ceiling is real. Subspace projection cannot fix it. SVD over Zipfian text is doomed. More raw text strictly degrades a flat substrate. These are not engineering bugs -- they are mathematical theorems. The substrate has the architectural capacity; the assumption that broke is "the substrate should learn grammar from raw text during the forward pass." Frontier LLMs only bypass Gold's Theorem of Language Identification in the Limit via backprop-driven global constraint-satisfaction over billions of parameters. We cannot do that. Therefore grammar MUST be discovered offline during sleep cycles via anti-unification across stored pointer chains. Sleep is not optional optimization -- it is where the learning happens.

The Attention-Resonator is the right tool wrongly applied to analogy. It is the right tool correctly applied to recall cleanup at generation. The VS-FSM is the right primitive for state-dependent transition. Abductive schema induction via the existing `schema_inducer.anti_unify` is the right primitive for structural discovery. Together they form the Day 67 compositional stack. The substrate has not yet generated coherent royal sentences from `the king` -- but it has discovered SVO structure from raw text, without supervision, in a single sleep cycle, on 17 sentences. That was not possible on Day 65.

The most important architectural decision of the day was made by Prince in one line: *"no one learns in single shot. I can't learn on single shot. what if we do that?"* The answer was yes. AWAKE = record. SLEEP = learn. GENERATE = walk attractors. The 192 MB constraint is no longer a bottleneck. The cosine ceiling is no longer a wall. The path is now: scale the corpus, scale the sleep cycles, watch the schema set densify, watch the abstract attractors deepen. The moonshot ([[concept-absorption-day63]] + [[universal-weight-generator]]) is no longer abstract architecture. It is engineering work.

Appendix A -- New inventions shipped Day 66-67

Out of the post-Day-65 inventory (88 cognition modules + 288 ikigai.py classes via Pack 216 bridge = 376 callable primitives), Day 66-67 added the following genuine novel inventions:

Tier A (substrate / addressing)

21. **schema_next role** (Pack 226, NEW Day 67) -- abstract grammatical state transitions stored as role-bound writes in the existing relational bank. The substrate's first explicit grammar channel. One phasor anchor; no bank growth.

Tier B (sleep + abstraction)

22. **Sleep-Sculpted Attractor Substrate (SSAS) attempt** (Pack 223, ATTEMPTED Day 66-67) -- combined attractor-distance measurement, basin merge/split, SVD principal-direction discovery, auto-labeling, concentrated reinforcement into a single sleep primitive. Empirically failed at 20% ceiling due to Zipfian SVD hijack and mass-distribution asymmetry. Theoretical attempt documented in `history.md` and `reference_research_2026_06_06.md`.
23. **lift_to_abstract -- isa-parent transition replication** (Pack 225, NEW Day 67) -- sleep-time substrate refinement: for every `(a, next, b)` concrete transition, write `(isa_parent(a), next, b)` and `(a, next, isa_parent(b))` and `(isa_parent(a), next, isa_parent(b))`. Concrete + abstract levels co-exist in same role. Generation queries blend both.
24. **induce_schemas -- abductive offline schema crystallization** (Pack 226, NEW Day 67) -- collects chains, abstracts via isa, groups by length, anti-unifies within groups via the existing `schema_inducer.anti_unify`, crystallizes schemas as `(a, schema_next, b)` transitions for consecutive non-SLOT positions. The Day 67 architectural commitment shipped as code.

Tier C (decoding)

25. **Attention-Resonator recall cleanup** (Pack 224, NEW Day 67) -- iterative continuous Hopfield log-sum-exp energy update over a candidate codebook. $r_{t+1} = \text{momentum} * r_t + (1 - \text{momentum}) * \text{softmax}(\text{beta} * K @ \text{conj}(r_t)) @ K$. Renorm at each step. Sharpens recall scores 1.4-2.2x at generation step. Wrong tool for analogy arithmetic (collapses multi-direction blend). Right tool for next-token prediction.

Tier D (generation)

26. **VS-FSM autoregression** (Pack 225, NEW Day 67) -- `VSFiniteStateMachine.generate(seed, max_tokens)` walks the substrate's `next` and `next2` attractors via `step` with optional trigram conditioning. Each step uses `resonator_recall` for candidate cleanup. Native API on `IkigaiOrganism` via `org.fsm_generate`.
27. **Schema-aware autoregression** (Pack 226, NEW Day 67) -- `VSFiniteStateMachine.generate_via_schema(seed, max_tokens)` interleaves abstract `schema_next` attractor lookup with concrete `next` attractor child resolution via `_isa_inverse`. Walks grammatical structure rather than bigram frequency. Native API: `org.fsm_generate_schema`.

Tier E (persistence + bridge)

28. **_gather_wired_state / _apply_wired_state -- 21-module pickle persistence** (Pack 220, NEW Day 66) -
- wraps every wired Pack 210-217 module's state into the universal `.ikg` via `npz pack220_keys / pack220_vals` arrays. Save 33 MB / 1.4s, load 0.7s. EVERY artifact survives process death. Bridge classes excluded intentionally (reload fresh from `ikigai.py` bridge).

29. **Frame routing in non-LLM reads + ToM agent_names()** fix (Pack 219, NEW Day 66) -- frame routing fires from regular `org.read_organism(text)` calls; ToM `agent_names()` method-call bug fixed (previous `self.tom = TheoryOfMindSandbox(d=d)` overwriting earlier `self.tom.add_agent('default')`).
30. **self_modifying_refiner.promote_check_wire** (Pack 218, NEW Day 66) -- refinement layer joins `sleep_consolidate` pipeline; substrate self-modification under sleep.

Methodological inventions (Tier F)

31. **The architectural commitment: drop one-shot learning** (Day 67 `mindset.md` rule 5) -- explicit decision that the organism learns in multi-pass cycles, with AWAKE phase recording only and SLEEP phase as the learning phase. Frees the substrate from Gold's Theorem intractability of online grammar induction. The most important architectural decision since the Day 57 flat-memory pivot.
32. **Memory consolidation 26 -> 8** (Day 67) -- collapsing 67 days of incremental memory files into a clean index + 4 thematic files + 2 references + the active plan. Reduces context drag on every session.

Appendix B -- The cosine ceiling, formally

The `renorm` step in `ConceptSynthesizer.build` produces unit-magnitude HVs. When K approximately-orthogonal HVs are bundled into one concept via `c = renorm(sum(role_recalls))`, the cosine similarity of `c` to any individual constituent is bounded by $1/\sqrt{K}$ with high probability. A concept with 1 strong property direction (e.g., `animal`) has cosine ~ 1.0 to that direction. A concept with 5 strong property directions (e.g., `kitten` tagged with `cat`, `baby`, `small`, `young`, `feline`) has cosine $\sim 1/\sqrt{5} \sim 0.447$ to any one.

In analogy arithmetic $V = \text{concept}(\text{cat}) - \text{concept}(\text{adult}) + \text{concept}(\text{baby})$, the dominant direction of V is whichever property mass is largest. Targets are ranked by cosine of V vs each candidate concept. Single-direction candidates (e.g., `animal`) fully align with their own direction in V (cosine close to 1). Rich-direction candidates (e.g., `kitten`) align partially with multiple directions in V but each only at $1/\sqrt{K}$. Single-direction loses to rich-direction only when V has very large mass in exactly one of the rich candidate's many directions -- which requires that the arithmetic vector cleanly picks out one direction. It does not for blends.

This is why Pack 158 hand-feeding works (hand-asserted properties are CONCENTRATED into one direction per word via 60x reinforcement, making each concept effectively single-direction along the relevant axis) and Pack 221-223 mining fails (mined properties are SPREAD across many directions, diluting each direction past $1/\sqrt{K}$ until the single-direction concepts dominate).

Subspace projection (Pack 223 v2) rotates the basis but preserves cosine ranking on the projected vectors. It does not fix the asymmetry. The math is rotation-invariant.

The Attention-Resonator (Pack 224) does break this for FACTOR DECOMPOSITION (find x, y, z given $S = x \text{ bound } y \text{ bound } z$) but does not help for additive blends. We confirmed this empirically: Pack 158 baseline 73%/100%/100%, Resonator iter=10 = 23%/39%/45%.

The escape from the ceiling, per the external research and our adopted commitment, is NOT to fix the cosine; it is to abandon arithmetic-based analogy in favor of state-machine generation. Generation is sequential -- one token at a time, conditional on current state. Each step's decoding problem IS a factor decomposition (recall the noisy `next` HV against the candidate codebook), exactly where Resonator helps. The substrate stops trying to do analogy arithmetic in concept space; it does grammar-walking in state space. Pack 225 and Pack 226 are the implementations.

This is the formal reason the moonshot is possible at 192 MB. Storage is not the bottleneck. The wrong tool was the bottleneck.